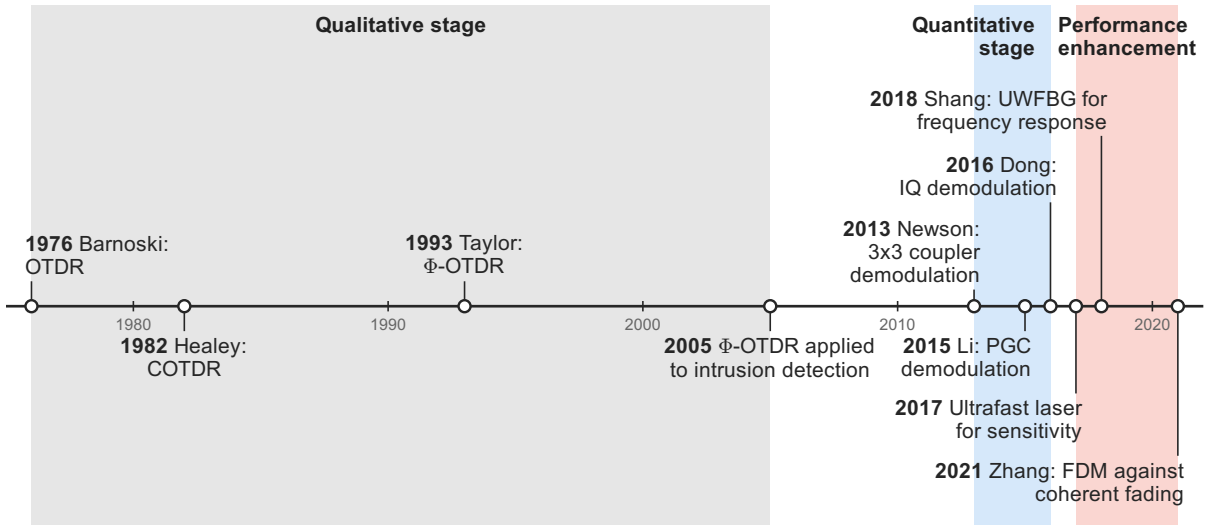
Qualitative stage
Quantitative stage
Performance enhancement
1976 Barnoski: OTDR
1982 Healey: COTDR
1993 Taylor: Φ-OTDR
2005 Φ-OTDR applied to intrusion detection
2013 Newson: 3x3 coupler demodulation
2015 Li: PGC demodulation
2016 Dong: IQ demodulation
2017 Ultrafast laser for sensitivity
2018 Shang: UWFBG for frequency response
2021 Zhang: FDM against coherent fading
1980
1990
2000
2010
2020